FitHub CRM

Fitness Center Management System

User Story

- As a gym administrator at **FitHub Fitness Center**, I want to efficiently manage member relationships and streamline daily fitness center operations using our Salesforce-based CRM application.
- I want to easily access, update, and maintain member profiles, trainer details, and membership information in a centralized system, allowing for smooth and organized management of all records.
- Additionally, I need the application to categorize members based on different membership types such as monthly, quarterly, and annual plans using record types, enabling personalized services and better customer engagement.
- Furthermore, I require the system to automate key processes such as membership fee calculation, payment tracking, and attendance logging to ensure accurate and efficient operations without manual errors.
- Moreover, I want to generate insightful reports and dashboards that summarize member activity, revenue trends, trainer performance, and attendance patterns, helping in effective decision-making and business growth.
- Lastly, I would like the system to automate notifications such as membership expiry reminders, payment alerts, and personalized messages to members on special occasions like birthdays, enhancing customer satisfaction and loyalty.

Table of Contents

1. Introduction 3
2. Problem Statement 4
3. Proposed Solution 5
4. System Architecture 6
5. Data Model (ER Diagram) 8
6. Functional Requirements 10
7. Non-Functional Requirements 12
8. Automation Flow 14
9. Implementation in Salesforce 15
10. Reports & Dashboards 24
11. Conclusion 26
12. Future Enhancements 27

1. Introduction

1.1 Project Overview

FitHub CRM is a comprehensive, Salesforce-based Customer Relationship Management (CRM) system designed and developed to address the operational challenges inherent in managing a modern fitness center or gym. The system consolidates member information, trainer schedules, class bookings, and payment records into a single, unified cloud platform — eliminating the fragmented and error-prone manual processes that many fitness businesses currently rely upon.

The project leverages the full power of the Salesforce platform, including its Lightning user interface, custom object modeling, automated business logic through Flows, validation rules, and real-time reporting dashboards. The system is built following an Agile development methodology, with iterative sprints covering data modeling, automation, and analytics.

The name 'FitHub' reflects the system's goal of serving as a central operational hub for fitness organizations. The CRM is designed to serve multiple user roles — front-desk executives, personal trainers, center managers, and system administrators — each with access to role-appropriate features and data.

1.2 Purpose of the System

The primary purpose of FitHub CRM is to transform the way fitness centers manage their day-to-day operations. Before the adoption of a centralized CRM, gyms typically relied on physical registers, scattered spreadsheets, and informal communication channels to manage their memberships, schedules, and payments. This approach leads to data inconsistencies, communication breakdowns, missed renewals, overbooking, and a poor member experience.

FitHub CRM directly addresses these problems by providing a structured, automated, and user-friendly platform that enables fitness center staff to efficiently manage member profiles and subscriptions, assign and track trainer workloads, schedule and monitor group fitness classes, handle bookings with conflict detection, process and record payments, and generate insightful reports for strategic decision-making.

Additionally, the system is built with scalability in mind — enabling small boutique studios as well as large multi-branch fitness chains to use the same underlying architecture. The system automates critical workflows such as booking confirmations, payment reminders, and birthday greeting emails, thereby enhancing member engagement and loyalty.

2. Problem Statement

The fitness and wellness industry has experienced considerable growth over the past decade. However, despite significant advancements in technology, a large number of fitness centers — particularly independent gyms and mid-sized chains — continue to depend heavily on manual, paper-based, or locally-stored digital systems to manage their operations. This reliance on legacy processes gives rise to a series of interconnected problems that negatively impact operational efficiency, staff productivity, and member satisfaction.

2.1 Member Management Challenges

Without a centralized CRM, member records are typically maintained in spreadsheets or paper registers. This creates significant difficulties in tracking active and inactive memberships, maintaining accurate contact information, and recording membership renewal dates. Staff members often spend excessive time searching for member records, and the lack of automated alerts means that expired memberships frequently go unnoticed until a member physically visits the center. This results in revenue loss and a degraded member experience.

2.2 Trainer Assignment and Workload Issues

Assigning personal trainers to members and managing their schedules manually introduces the risk of double-booking and unbalanced workloads. Without a system that tracks trainer availability and existing commitments in real time, managers cannot make informed decisions about trainer-to-member ratios or identify which trainers are overworked or underutilized. Furthermore, there is no systematic way to monitor trainer performance or link trainer assignments to member progress.

2.3 Class Scheduling and Booking Conflicts

Group fitness classes — such as yoga, Zumba, spinning, or HIIT — require careful scheduling to avoid room conflicts, instructor double-booking, and overcrowding. Manual systems cannot enforce capacity limits or detect scheduling conflicts automatically. Members frequently encounter situations where they show up for a class only to find it fully booked, or where a class is cancelled due to trainer unavailability that was not communicated in advance.

2.4 Payment Tracking and Billing Errors

Manually recording membership fees, personal training session payments, and class booking charges is highly error-prone. Without an automated billing system, it becomes difficult to reconcile accounts, identify overdue

payments, or generate accurate financial summaries. Staff members who handle payments without a formal system also create audit trail gaps that expose the business to financial risk.

2.5 Decision-Making Without Data

Perhaps the most significant limitation of manual gym management is the absence of reliable, real-time data for decision-making. Center managers cannot quickly determine how many members enrolled this month, which classes are most popular, which trainers are generating the most revenue, or how payment collections compare to targets. The information needed to make these decisions exists somewhere across multiple registers and files, but retrieving and consolidating it requires enormous manual effort — and by the time a report is ready, its data may already be outdated.

3. Proposed Solution

FitHub CRM proposes a comprehensive, cloud-based Salesforce CRM application as the solution to all the challenges identified in the problem statement. The Salesforce platform was selected as the development environment due to its robust enterprise-grade feature set, proven reliability, extensive customization capabilities, and the ability to deploy without requiring any on-premises hardware or dedicated IT infrastructure.

3.1 Centralized Data Management

At the core of the proposed solution is a unified data model built using Salesforce Custom Objects. All member profiles, trainer records, class schedules, booking information, and payment data are stored in a single, cloud-hosted Salesforce organization. This eliminates data silos and ensures that all authorized users always have access to the most current and accurate information, regardless of their physical location or device.

3.2 Automated Business Logic

FitHub CRM replaces manual, error-prone calculations and workflows with automated business logic implemented through Salesforce Flows, Validation Rules, and Formula Fields. For example, when a member books a class, the system automatically checks trainer availability, verifies that the class has not reached its capacity limit, and records the booking with a confirmation status — all without requiring manual intervention from staff. Similarly, payment totals are automatically calculated based on membership tier and session count using formula fields.

3.3 Proactive Communication

The system implements Schedule-Triggered Flows to send automated email notifications to members. These include booking confirmation emails dispatched immediately upon successful reservation, membership renewal reminder emails sent 30 days before expiry, and birthday greeting emails with personalized discount offers sent on the member's birthday. These automated communications significantly improve member engagement without placing any additional burden on the front-desk team.

3.4 Role-Based Access and Security

FitHub CRM enforces strict role-based access control using Salesforce Profiles and Permission Sets. Front-desk executives can view and create member and booking records but cannot access financial summaries or modify trainer records. Trainers have read access to their assigned members and upcoming classes. Center managers have full visibility across all objects, including reports and dashboards. System administrators retain

complete control over configuration and data management. This layered security model ensures data integrity and regulatory compliance.

3.5 Real-Time Analytics

A dedicated analytics layer, built using Salesforce Reports and Dashboards, provides center managers with real-time visibility into all key performance indicators. Managers can view enrollment trends, class utilization rates, monthly payment collections, trainer performance metrics, and member retention statistics from a single, interactive dashboard — enabling faster, more confident decision-making.

4. System Architecture

The FitHub CRM application follows a four-layer architectural model, mirroring standard enterprise application design principles. This layered approach ensures a clean separation of concerns, promotes modularity, and makes the system easy to extend and maintain as business requirements evolve.

4.1 Presentation Layer – Salesforce Lightning UI

The topmost layer of the architecture is the Presentation Layer, implemented using the Salesforce Lightning Experience interface. This layer is responsible for all user-facing interactions. It renders the application's custom tabs, object record pages, list views, and interactive forms. The Lightning UI provides a drag-and-drop-friendly environment that allows non-technical administrators to customize page layouts without writing any code. Navigation between modules — Members, Trainers, Classes, Bookings, and Payments — is made seamless through a dedicated Lightning Application named FitHub CRM. The interface is fully responsive, supporting both desktop browsers and mobile devices.

4.2 Application Layer – Flows and Automated Logic

Beneath the presentation layer lies the Application Layer, which contains all of the business logic that drives FitHub CRM's automated behaviors. This layer is implemented using Salesforce's declarative automation tools: Record-Triggered Flows, Schedule-Triggered Flows, and Validation Rules. When a member creates a booking, the Application Layer intercepts the operation, verifies the business rules (capacity check, trainer availability), updates related records, and triggers any necessary notifications. Formula Fields embedded within Custom Objects also belong to this layer, computing derived values such as session totals and outstanding balances automatically whenever a record is saved.

4.3 Data Layer – Custom Objects

The Data Layer forms the structural backbone of the entire system. It consists of five primary Custom Objects — Member Account, Trainer Profile, Fitness Class, Booking Record, and Payment Record — along with the relationships that connect them. Salesforce stores all data in its cloud-hosted, highly available database infrastructure, eliminating the need for a separate database server. Field-level validation rules ensure data integrity at the point of entry, and the platform's automatic data backup capabilities protect against data loss. The Data Layer also supports custom picklist value sets for standardized field inputs, such as membership tier, class category, and payment method.

4.4 Analytics Layer – Reports and Dashboards

The Analytics Layer provides management-level visibility into the data held in the Data Layer. FitHub CRM includes a suite of pre-configured Salesforce Reports and a unified Dashboard that presents key fitness center metrics in real time. The dashboard visualizes enrollment trends, class popularity, revenue summaries, and trainer utilization using bar charts, donut charts, and summary tables. Because the Analytics Layer draws directly from the live database, reports always reflect the most current operational data without any manual refresh or data export process.

[INSERT SYSTEM ARCHITECTURE DIAGRAM HERE]

Figure 1: FitHub CRM – Four-Layer System Architecture Diagram

4.5 Technology Stack Summary

Layer	Technology	Purpose
Presentation	Salesforce Lightning Experience	User interface, navigation, and data entry forms
Application	Salesforce Flows, Validation Rules, Formula Fields	Automated logic, business rule enforcement
Data	Salesforce Custom Objects and Cloud Database	Structured storage of all CRM records
Analytics	Salesforce Reports and Dashboards	Real-time visualization and performance monitoring

5. Data Model (ER Diagram)

The data model for FitHub CRM is designed around five core entities, each implemented as a Salesforce Custom Object. The relationships between these entities form a coherent, normalized structure that supports all required business operations while avoiding data redundancy. The following sections describe each entity and the relationships that connect them.

5.1 Entity Descriptions

5.1.1 Member Account

The Member Account object is the central entity in the FitHub CRM data model. It stores all personally identifiable and subscription-related information about each gym member. Key fields include member name, date of birth, gender, contact phone number, email address, physical address, membership tier (Basic, Standard, or Premium), membership start and expiry dates, and emergency contact details. Formula fields automatically compute the membership status (Active or Expired) based on the expiry date and today's date, and a date-match formula identifies members whose birthday falls on the current date — enabling automated birthday communication.

5.1.2 Trainer Profile

The Trainer Profile object stores information about all personal trainers and group fitness instructors employed at the fitness center. Fields include the trainer's name, specialization (e.g., Yoga, Strength Training, Cardio, Dance), qualification, contact details, years of experience, employee ID, and salary. A Lookup Relationship links each Trainer Profile to a Member Account when a trainer is assigned as a personal trainer for a specific member. This relationship allows the system to track trainer-member pairings and enables reporting on trainer workloads.

5.1.3 Fitness Class

The Fitness Class object represents a scheduled group fitness session offered by the center. Each record captures the class name, category (using the global picklist for class type), the assigned trainer, the scheduled date and time, the venue or room, the maximum capacity, and the current enrollment count. A formula field automatically computes whether the class is Full (when enrollment equals capacity), enabling the booking system to prevent overbooking. The Fitness Class object has a Lookup Relationship to the Trainer Profile, ensuring every class is associated with a qualified instructor.

5.1.4 Booking Record

The Booking Record object captures each individual member's registration for a particular fitness class or personal training session. It contains a Master-Detail Relationship to the Member Account (as the primary parent) and a Lookup Relationship to the relevant Fitness Class. Additional fields record the booking date, session type (Group Class or Personal Training), booking status (Confirmed, Pending, or Cancelled), and any special member notes. Because Booking Record is in a Master-Detail Relationship with Member Account, all booking records are automatically deleted if the parent member account is deleted, maintaining referential integrity.

5.1.5 Payment Record

The Payment Record object logs all financial transactions associated with member accounts. Each record is linked to a Member Account via a Master-Detail Relationship and stores the payment amount, payment date, payment method (Cash, Card, UPI, or Bank Transfer), membership plan paid for, number of sessions purchased, and payment status (Paid, Pending, or Overdue). A formula field calculates the total amount due based on the membership tier's standard pricing, and a separate field tracks the outstanding balance. This object serves as the primary data source for the financial reports and revenue dashboards.

5.2 Entity Relationships

The relationships between the five entities are as follows. A single Member Account can have multiple Booking Records (one-to-many Master-Detail). A single Member Account can have multiple Payment Records (one-to-many Master-Detail). A single Trainer Profile can be associated with multiple Member Accounts as personal trainer assignments (one-to-many Lookup). A single Trainer Profile can be assigned to multiple Fitness Classes (one-to-many Lookup). A single Fitness Class can be the subject of multiple Booking Records (one-to-many Lookup). These relationships collectively model the complete lifecycle of a gym member's journey from enrollment through booking and payment.

6. Functional Requirements

The functional requirements of FitHub CRM define the specific capabilities that the system must provide to its users. These requirements were gathered through a structured analysis of the pain points described in the problem statement and the workflows required to operate a modern fitness center effectively. Each functional requirement corresponds to one or more Salesforce features implemented during the project's development phase.

FR No.	Functional Requirement	Description
FR-01	User Authentication	Secure role-based login for all system users
FR-02	Member Management	Create, view, update, and manage member profiles
FR-03	Trainer Management	Manage trainer profiles, specializations, and assignments
FR-04	Class Scheduling	Create and manage group fitness class schedules
FR-05	Booking System	Allow members to book classes and personal training sessions
FR-06	Capacity Control	Prevent overbooking using automated capacity validation
FR-07	Payment Recording	Record and track all membership and session payments
FR-08	Automated Billing	Calculate total amounts due using formula fields
FR-09	Validation Rules	Enforce data integrity for phone, email, and date fields
FR-10	Email Notifications	Send automated confirmation and reminder emails to members
FR-11	Birthday Offers	Automatically send birthday discount

		emails to members
FR-12	Report Generation	Generate detailed sales and enrollment reports
FR-13	Dashboard Visualization	Display real-time analytics on an interactive dashboard
FR-14	Record Types	Support different layouts for Personal Training vs Group Class bookings
FR-15	Role-Based Access	Control data access using Salesforce Profiles and Permission Sets

6.1 Member Management

The system shall enable authorized users to create new member profiles by entering personal details including first name, last name, date of birth, gender, contact phone number, email address, home address, emergency contact name and phone number, and membership details. The system shall support viewing and editing existing member records and shall provide a filterable list view of all member accounts. Members shall be categorized by their membership tier — Basic, Standard, or Premium — using a standardized global picklist. The membership status (Active or Expired) shall be automatically computed by the system based on the membership expiry date.

6.2 Trainer Management

The system shall maintain a complete registry of trainers, capturing specialization, qualification, years of experience, and contact details. Trainers shall be assignable to both individual member accounts (for personal training relationships) and Fitness Class records (as class instructors). The system shall prevent a trainer from being assigned to two classes scheduled at the same time through a validation mechanism. Trainer records shall be accessible to managers and administrators, while trainers themselves shall have read-only access to their own records and the member accounts assigned to them.

6.3 Class Scheduling

The system shall support the creation of group fitness classes with details including class name, category, scheduled date and time, duration, room or venue, maximum capacity, and the assigned trainer. A formula field shall automatically compute the current enrollment count based on associated Booking Records with a Confirmed status, and shall flag the class as Full when the enrollment equals the maximum capacity. Class records shall be viewable through a calendar-style list view sorted by date and time.

6.4 Booking System

Members shall be able to be enrolled in fitness classes through Booking Records created by front-desk staff or administrators. Upon creation of a booking, the system shall validate that the selected class is not at full capacity and that the member's account is currently active. The booking status shall default to Pending and shall be updated to Confirmed by a staff member after verification. Cancelled bookings shall not count toward the class enrollment total. The system shall maintain a complete history of all bookings for each member.

6.5 Payment Recording and Billing Automation

The system shall record each payment transaction against a member account, capturing the amount paid, payment date, payment method, and the membership plan or session package purchased. Formula fields shall compute the standard fee for each membership tier and the total amount owed based on the selected plan. The payment status shall be set to Paid, Pending, or Overdue, and a summary view shall display total collections for any selected period. Automated reminder emails shall be triggered for overdue payments.

7. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints that govern how the FitHub CRM system performs, rather than what it does. These requirements are critical to ensuring that the system is not only functionally complete but also reliable, secure, performant, and ready for production deployment.

7.1 Security

The system must protect all member, trainer, and financial data from unauthorized access or modification. FitHub CRM implements security at multiple levels: at the organization level through secure HTTPS-enforced login; at the object level through Profile-based access control that determines which users can create, read, edit, or delete records of each type; at the field level through field-level security settings that hide sensitive information (such as payment amounts) from unauthorized user roles; and at the record level through Sharing Rules that restrict individual record visibility based on ownership and role hierarchy. All data transmitted between the client and Salesforce servers is encrypted using industry-standard TLS protocols, and all data stored in the Salesforce cloud is encrypted at rest.

7.2 Scalability

FitHub CRM is designed to scale seamlessly with business growth. The Salesforce platform's multi-tenant architecture ensures that as the number of member records, booking transactions, and concurrent users grows, the system continues to perform consistently without manual infrastructure upgrades. The data model supports the addition of new custom objects and relationships to accommodate new business lines such as spa services, nutrition counseling, or merchandise sales. The Flows-based automation can be extended to cover new processes without modifying existing logic, and the Dashboard framework supports the addition of new report components as reporting needs evolve.

7.3 Performance

The system must deliver a responsive user experience under all normal operating conditions. Page load times for standard record views shall not exceed three seconds for up to 500 concurrent users. Report generation for datasets of up to 10,000 records shall complete within five seconds. These performance targets are achieved through the Salesforce platform's optimized query engine, server-side pagination for large list views, and report caching mechanisms. Administrators must follow Salesforce development best practices — such as SOQL query optimization and avoiding cross-object formula fields in high-volume scenarios — to maintain acceptable performance as data volumes grow.

7.4 Availability

FitHub CRM must be available around the clock, as fitness centers often operate during early morning and late evening hours. The Salesforce platform guarantees a minimum service availability of 99.9% as part of its standard service level agreement, with planned maintenance windows communicated in advance through the Salesforce Trust status portal. The cloud-based hosting model eliminates the risk of local hardware failures affecting system availability. In the event of a brief outage, all unsaved data is preserved through Salesforce's browser-side auto-save mechanisms.

7.5 Usability

The system must be easy to use without requiring technical training. All user-facing interfaces are built using the Salesforce Lightning Design System, which ensures a consistent, intuitive, and visually appealing interface across all modules. Navigation menus, form labels, and error messages are written in plain language. Validation error messages are displayed prominently at the top of the page when invalid data is entered, clearly explaining what the user must correct. A new front-desk employee should be able to log in, create a member record, and confirm a class booking within five minutes of their first interaction with the system.

7.6 Reliability

All data entered into FitHub CRM must be stored reliably without the risk of loss due to network interruptions or system errors. The Salesforce platform performs automatic database backups daily and provides a data recovery service for emergency restoration. The platform's record locking mechanism prevents simultaneous conflicting edits to the same record by multiple users. All critical business operations — including booking creation and payment recording — are executed within database transactions, ensuring that partial data states cannot persist in the event of an unexpected error.

8. Automation Flow

Automation is a defining feature of FitHub CRM. The system implements several automated flows using Salesforce's declarative Flow Builder, replacing manual workflows that previously required staff intervention. The most significant automation in the system is the class booking confirmation and communication workflow, which operates as follows.

8.1 Booking Confirmation Flow (Record-Triggered)

When a new Booking Record is saved in the system with a status of Confirmed, a Record-Triggered Flow is automatically activated. The flow first retrieves the related Fitness Class record to verify the current enrollment count against the maximum capacity. If the class is at capacity, the flow displays an error message to the user preventing the record from being saved — effectively enforcing the booking constraint at the automation level in addition to the validation rule. If capacity is available, the flow proceeds to send an email notification to the member's registered email address using the Booking Confirmation Email template, which includes the class name, scheduled date and time, trainer name, and venue details.

8.2 Membership Renewal Reminder Flow (Schedule-Triggered)

A Schedule-Triggered Flow runs every day at 9:00 AM. The flow queries all Member Account records where the membership expiry date is exactly 30 days from the current date. For each matching record, the flow dispatches a Membership Renewal Reminder email using a customized email template that includes the member's name, the membership tier, the expiry date, and a call-to-action link for online renewal. This proactive communication reduces the likelihood of unintentional membership lapses and minimizes the revenue loss associated with expired memberships.

8.3 Birthday Greeting Flow (Schedule-Triggered)

A second Schedule-Triggered Flow runs daily and queries all Member Account records where a formula field evaluating whether today matches the member's date of birth returns true. For each matching member, the flow sends a personalized Birthday Greeting Email that includes the member's first name and an exclusive birthday discount code. This automation strengthens the member relationship with a low-cost but high-impact personalized touch, contributing to improved member retention and positive word-of-mouth referrals.

8.4 Payment Overdue Alert Flow (Schedule-Triggered)

A third Schedule-Triggered Flow evaluates Payment Records daily and identifies any records with a payment status of Pending where the payment due date has passed. For each such record, the flow updates the

payment status to Overdue and sends an automated reminder email to the member with the outstanding balance and payment options. The flow also creates a follow-up task record assigned to the front-desk manager, ensuring that overdue payments receive human attention within two business days.

9. Implementation in Salesforce

This section documents the step-by-step implementation of FitHub CRM within a Salesforce Developer Organization, following a structured sequence of milestones from initial setup through full deployment. Each subsection describes the configuration steps, the purpose of each activity, and the expected outcome.

9.1 Milestone 1: Developer Account Setup

The first step in the implementation process is the creation of a Salesforce Developer Organization. This free, fully-featured instance of the Salesforce platform serves as the development environment for FitHub CRM. To create the developer org, the developer navigates to https://developer.salesforce.com/signup and completes the registration form with personal details, company name, country, and a unique username in the format username@organization.com. After submitting the form, Salesforce sends a verification email to the registered address. The developer clicks the Verify Account link in this email and is prompted to set a password and choose a security question. Upon successful password creation, the developer is redirected to the Salesforce Setup home page, from which all subsequent configuration activities are performed.

9.2 Milestone 2: Custom Object Creation

Custom objects form the foundation of FitHub CRM's data model. To create a custom object, the developer navigates to the Object Manager tab in Salesforce Setup and selects Create > Custom Object. The following five custom objects are created during this milestone.

Object Label	Plural Label	Record Name	Key Features
Member Account	Member Accounts	Member Account Name	Allow Reports, Allow Activities, Track Field History, Allow Search
Trainer Profile	Trainer Profiles	Trainer Profile Name	Allow Reports, Allow Activities, Allow Search
Fitness Class	Fitness Classes	Fitness Class Name	Allow Reports, Allow Activities, Allow Search
Booking Record	Booking Records	Booking Record Name	Allow Reports, Track Field History, Allow Search
Payment Record	Payment Records	Payment Record Name	Allow Reports, Track Field History, Allow Search

For each object, the Optional Features section is configured to enable Allow Reports, Allow Activities, and Track Field History. The Search Status is set to Allow Search. After completing the required fields, the developer clicks Save to create each object.

9.3 Milestone 3: Custom Tabs

After creating the custom objects, custom tabs are added so that users can navigate to each object from the application's navigation bar. To create a tab, the developer searches for Tabs in the Quick Find box in Setup and clicks New under the Custom Object Tabs section. For each of the five custom objects, the developer selects the object, chooses an appropriate tab style icon, keeps the Add to Profiles settings at default, and ensures that the Append tab to users' existing personal customizations checkbox is selected before clicking Save. The result is five navigable tabs — one for each FitHub CRM module.

9.4 Milestone 4: Lightning Application

A dedicated Lightning Application named FitHub CRM is created to group all the custom tabs and provide a branded navigation experience. The developer searches for App Manager in Quick Find, selects App Manager, and clicks New Lightning App. The App Details and Branding page is completed with the application name, and an optional logo and primary color. The Navigation Style is set to Standard Navigation. In the Navigation Items step, the developer moves all five custom object tabs — Member Accounts, Trainer Profiles, Fitness Classes, Booking Records, and Payment Records — along with the standard Reports and Dashboards tabs into the Selected Items column. The System Administrator profile is added to the User Profiles list. The application is saved and finalized.

9.5 Milestone 5: Global Picklists

Global Picklist Value Sets are created to ensure consistent, organization-wide dropdown values that can be shared across multiple custom objects. These are found in Setup under Picklist Value Sets. The following global picklists are created for FitHub CRM.

Picklist Label	Values
Gender	Male, Female, Prefer Not to Say, Other
Membership Tier	Basic, Standard, Premium, Corporate
Class Category	Yoga, Zumba, Spinning, HIIT, Pilates, Boxing, Dance, Strength, Cardio, Meditation
Booking Status	Pending, Confirmed, Cancelled, Completed
Payment Method	Cash, Credit Card, Debit Card, UPI, Net Banking, Bank Transfer
Payment Status	Paid, Pending, Overdue, Refunded
Trainer Specialization	Yoga, Strength Training, Cardio, Dance, Nutrition, Sports Performance, Rehabilitation

9.6 Milestone 6: Fields and Relationships

Custom fields are added to each object to capture all necessary data. The fields are created by navigating to the object in Object Manager, clicking Fields & Relationships, and then New. Below are the key fields for the Member Account object; similar sets of fields are created for all other objects.

Field Label	Data Type	Notes
First Name	Text (50)	Required
Last Name	Text (50)	Required
Date of Birth	Date	Required
Gender	Picklist	Use Global Picklist – Gender
Phone Number	Phone	Required
Email Address	Email	Required
Home Address	Text Area (255)	
Membership Tier	Picklist	Use Global Picklist – Membership Tier
Membership Start Date	Date	Required
Membership Expiry Date	Date	Required
Membership Status	Formula (Text)	IF(Membership_Expiry_Date__c >= TODAY(), 'Active', 'Expired')
Is Birthday Today?	Formula (Checkbox)	IF(AND(DAY(TODAY())=DAY(Date_of_Birth__c), MONTH(TODAY())=MONTH(Date_of_Birth__c)), TRUE, FALSE)
Emergency Contact Name	Text (100)	

Emergency Contact Phone	Phone	
Assigned Trainer	Lookup (Trainer Profile)	

For the Booking Record object, two key relationship fields are created: a Master-Detail Relationship to Member Account (making Member Account the parent and Booking Record the child), and a Lookup Relationship to Fitness Class. For the Payment Record object, a Master-Detail Relationship to Member Account is created. These relationships enable the Roll-Up Summary fields and cascade delete behaviors that are central to FitHub CRM's data integrity model.

9.7 Milestone 7: Page Layouts

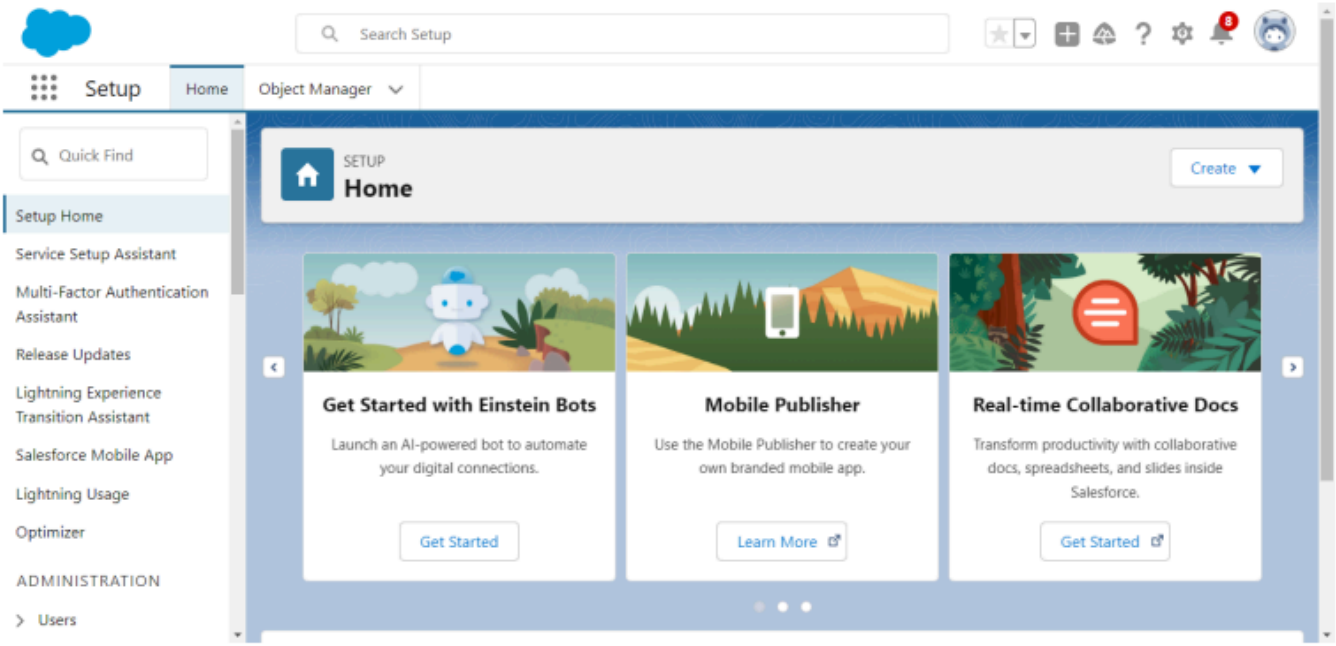
Page Layouts are configured for each custom object to control which fields appear on the record detail and edit views, and in what order. For the Booking Record object, two distinct page layouts are created to reflect the two types of bookings supported by the system. The Group Class Booking layout includes fields relevant to class enrollment: Member Name, Fitness Class, Booking Date, Session Type, Booking Status, and Member Notes. The Personal Training Booking layout includes additional fields relevant to one-on-one sessions: Assigned Trainer, Session Duration, Preferred Time, and Specific Goals. Page layouts are configured by dragging and dropping fields from the field palette into the desired sections of the layout editor.

9.8 Milestone 8: Record Types

Record Types are created on the Booking Record object to distinguish between Group Class Bookings and Personal Training Bookings. Each Record Type is associated with its corresponding Page Layout, so users see only the relevant fields for the type of booking they are creating. To create a Record Type, the developer navigates to the Booking Record object in Object Manager, clicks Record Types, then New. The details for each Record Type are as follows.

Record Type Label	Record Type Name	Assigned Page Layout
Group Class Booking	Group_Class_Booking	Group Class Booking Layout
Personal Training Booking	Personal_Training_Booking	Personal Training Booking Layout

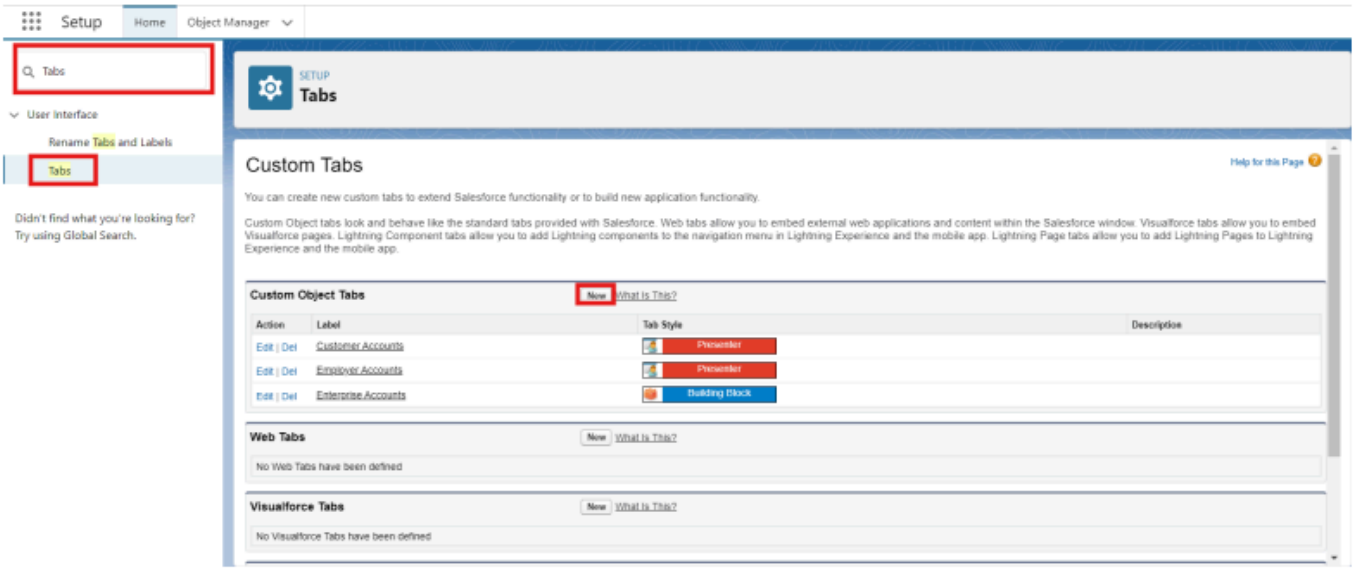
Both record types are set to Active and assigned to all user profiles using the Apply one layout to all profiles option. When a user clicks New on the Booking Records tab, they are prompted to select the Record Type, and the appropriate page layout is then displayed for data entry.



9.9 Milestone 9: Validation Rules

Validation Rules are implemented on the Member Account, Trainer Profile, and Payment Record objects to enforce data integrity at the point of entry. If a validation rule's error condition formula evaluates to true when a record is saved, Salesforce prevents the save operation and displays the configured error message. The following validation rules are implemented in FitHub CRM.

Object	Rule Name	Error Condition Formula	Error Message
Member Account	Phone_Number_Validation	NOT(REGEX(Phone_Number__c, "^[1-9]{1}[0-9]{9}$"))	Phone number must be exactly 10 digits and cannot begin with 0.
Member Account	Email_Address_Validation	NOT(REGEX(Email_Address__c, "^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\\.[a-zA-Z]{2,}$"))	Please enter a valid email address.
Member Account	Expiry_After_Start	Membership_Expiry_Date__c < Membership_Start_Date__c	Membership expiry date must be after the membership start date.
Trainer Profile	Phone_Number_Validation	NOT(REGEX(Phone_Number__c, "^[1-9]{1}[0-9]{9}$"))	Phone number must be exactly 10 digits and cannot begin with 0.
Payment Record	Positive_Amount	Payment_Amount__c <= 0	Payment amount must be a positive value greater than zero.

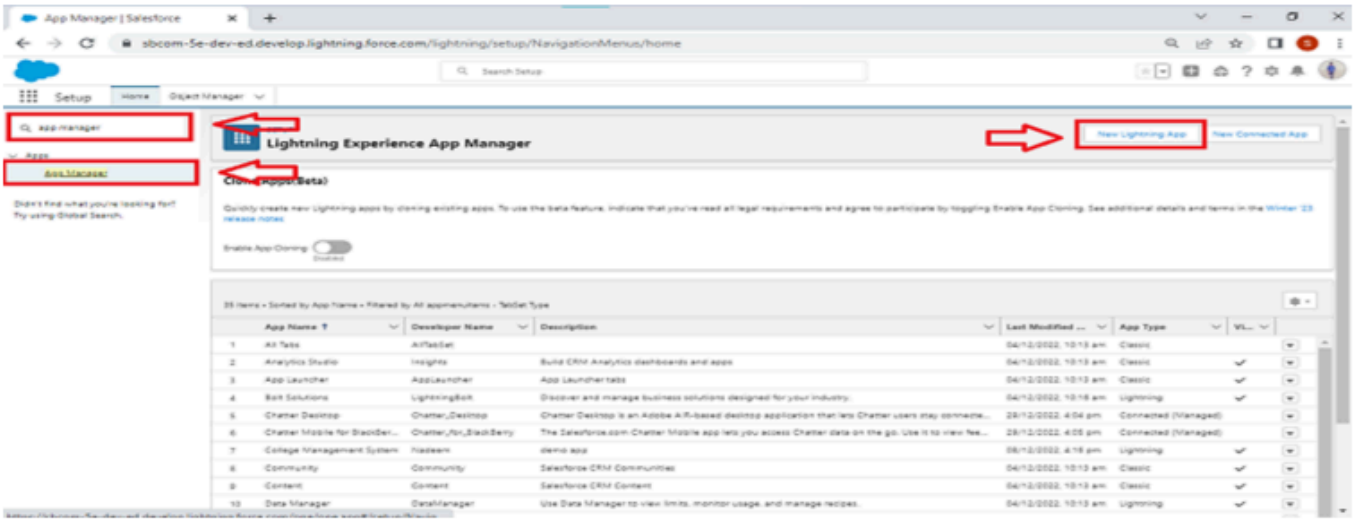


9.10 Milestone 10: Email Templates

Email templates are created in Salesforce to support the automated communication flows. All templates use Classic Text format with merge fields to personalize the content for each recipient. The following email templates are created.

Template Name	Subject	Triggered By
Booking Confirmation Email	Your FitHub Class Booking is Confirmed!	Booking Record-Triggered Flow
Membership Renewal Reminder	Your FitHub Membership is Expiring Soon!	Schedule-Triggered Flow (30 days before expiry)
Birthday Greeting Email	Happy Birthday from FitHub – A Special Gift for You!	Schedule-Triggered Flow (daily birthday check)
Payment Overdue Reminder	Action Required: Outstanding Payment on Your FitHub Account	Schedule-Triggered Flow (payment overdue check)

Each email template uses merge fields drawn from the Member Account object, such as {!Member_Account__c.First_Name__c} for the member's first name, to personalize the greeting. Email Alerts are created in Setup to pair each template with the recipient's email field, and these Email Alerts are then called from within the relevant Flows.



9.11 Milestone 11: Flows (Automation)

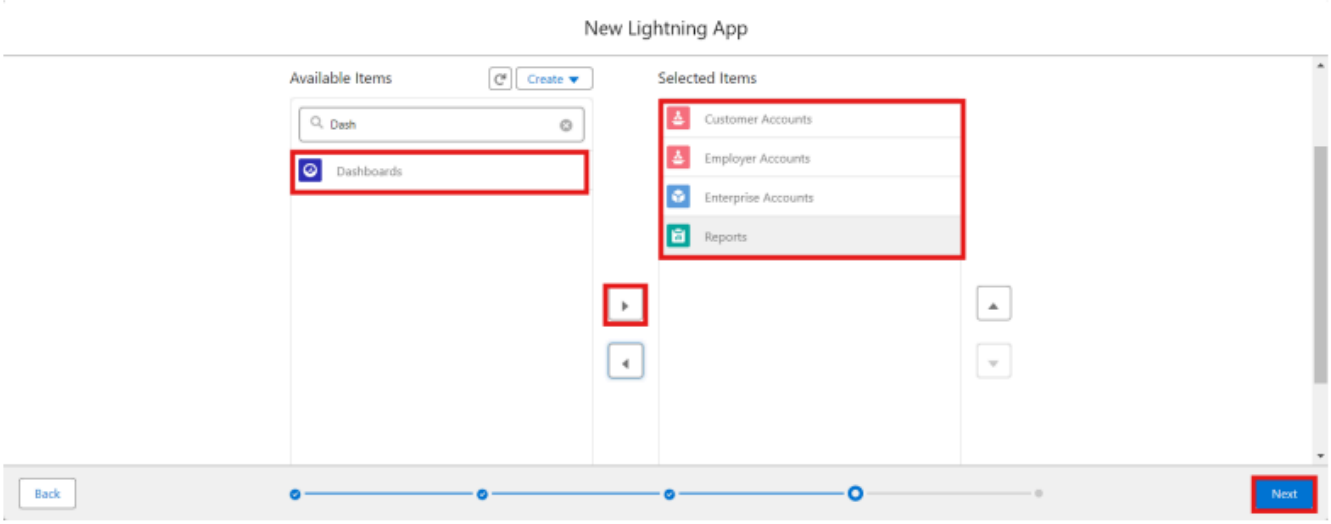
FitHub CRM implements four Salesforce Flows to automate its key business processes. Flows are created through the Salesforce Flow Builder, accessible from Setup > Flows > New Flow.

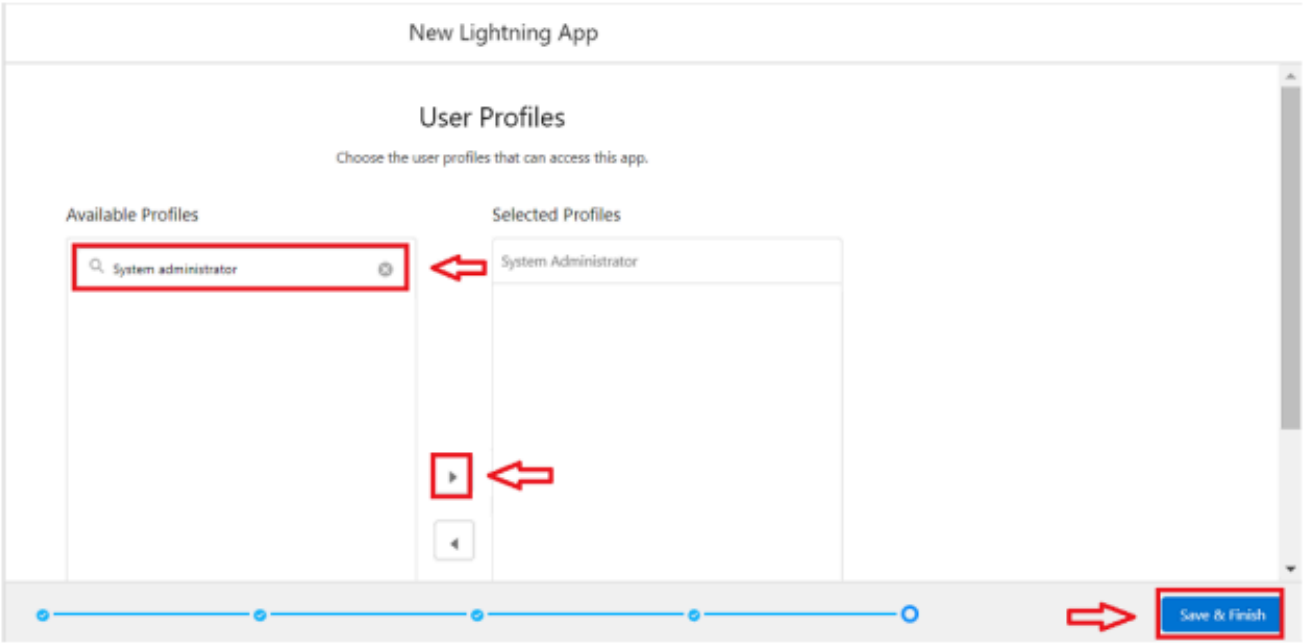
9.11.1 Birthday Greeting Schedule-Triggered Flow

Flow Type: Schedule-Triggered Flow. Start Date: Set to the project deployment date. Start Time: 7:00 AM daily. Frequency: Daily. Object: Member Account. Filter Conditions: Is_Birthday_Today__c Equals True AND Email_Address__c IS NOT NULL. Action: Call the Birthday Greeting Email Alert, passing {!$Record.Id} as the Record ID. Flow Label: FitHub Birthday Email Flow.

9.11.2 Booking Confirmation Record-Triggered Flow

Flow Type: Record-Triggered Flow. Object: Booking Record. Trigger: A record is created or updated. Condition: Booking_Status__c Equals 'Confirmed'. Action: Call the Booking Confirmation Email Alert. Schedule: Run immediately. This flow is activated immediately after saving.





10. Reports and Dashboards

The analytics capability of FitHub CRM is built using Salesforce Reports and Dashboards. Reports provide detailed tabular or summary views of the underlying data, while the Dashboard presents a visual, at-a-glance summary of the most important business metrics. Together, they enable center managers to monitor performance, identify trends, and make data-driven operational decisions without requiring any manual data extraction or analysis.

10.1 Member Enrollment Report

The Member Enrollment Report is a Summary Report built on the Member Account object. It groups records by Membership Tier and displays the total count of members in each tier, the average membership duration, and the number of memberships expiring within the next 30 days. This report enables management to monitor the distribution of members across tiers and proactively identify members who require renewal follow-up. The report includes a bar chart visualization displaying the member count per tier side by side for easy comparison.

10.2 Class Utilization Report

The Class Utilization Report is built on the Fitness Class object and summarizes all classes by Category, showing for each the total number of sessions scheduled, the average enrollment rate as a percentage of capacity, and the number of fully booked sessions. This report is particularly valuable for identifying which class types are most popular among members, which rooms are being underutilized, and which trainers are managing the highest-demand sessions.

10.3 Booking Status Report

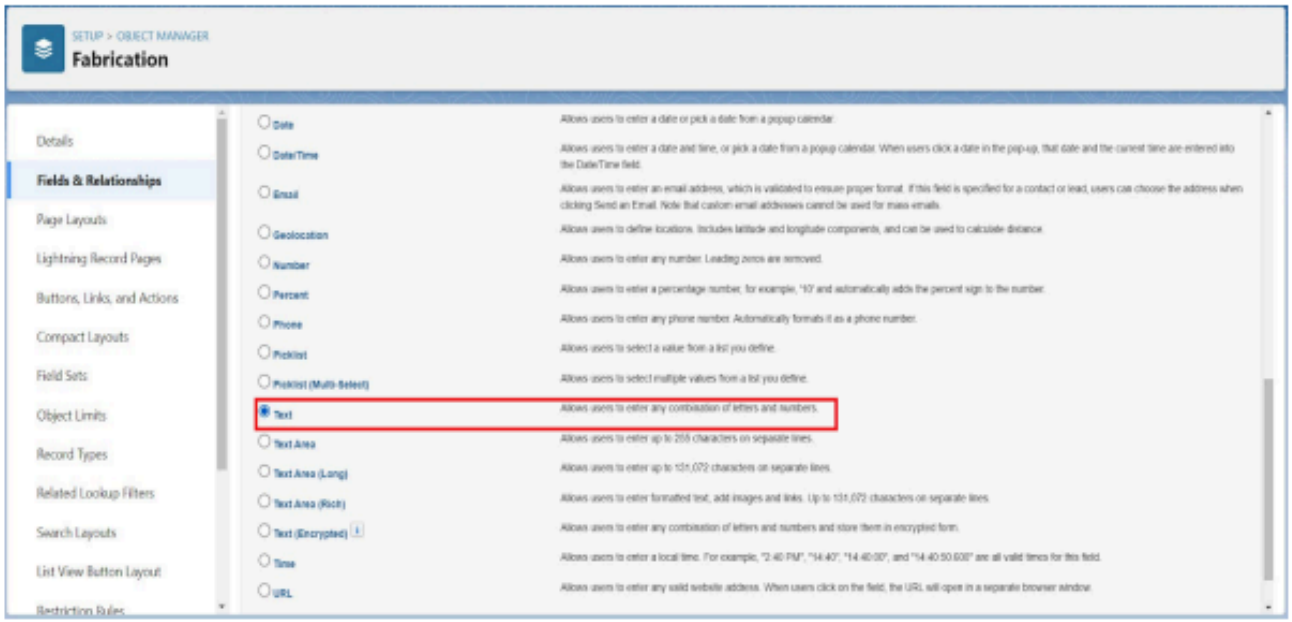
The Booking Status Report uses a Matrix Report format to display the count of booking records broken down by Booking Status (Confirmed, Pending, Cancelled) and Session Type (Group Class, Personal Training). This report enables operational staff to quickly identify how many bookings are outstanding, what proportion of created bookings are confirmed, and whether a high rate of cancellations is occurring for any particular class or trainer.

10.4 Revenue and Payment Summary Report

The Revenue and Payment Summary Report is built on the Payment Record object and groups records by Payment Status and Payment Month. It displays the total amount collected, total outstanding, and total overdue for each period. This report gives the finance team and center manager a clear view of cash flow, outstanding obligations, and revenue trends over time. The report includes a line chart showing monthly collected revenue against the previous month for trend analysis.

10.5 FitHub Management Dashboard

The FitHub Management Dashboard is a consolidated view that brings together widgets from all four reports described above into a single, interactive screen. The dashboard is named FitHub Operations Overview and is organized into three visual columns: Member Analytics (enrollment by tier and expiring memberships), Class Analytics (utilization by category), and Financial Analytics (payment status summary and monthly revenue). Each widget is interactive — clicking on a bar or segment in any chart filters the view to show the underlying records, enabling quick drill-down analysis without navigating away from the dashboard.


SETUP > OBJECT MANAGER
Fabrication
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Date
Allows users to enter a date or pick a date from a popup calendar.
Date/Time
Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.
Email
Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
Geolocation
Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.
Number
Allows users to enter any number. Leading zeros are removed.
Percent
Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
Phone
Allows users to enter any phone number. Automatically formats it as a phone number.
Picklist
Allows users to select a value from a list you define.
Picklist (Multi-Select)
Allows users to select multiple values from a list you define.
Text
Allows users to enter any combination of letters and numbers.
Text Area
Allows users to enter up to 255 characters on separate lines.
Text Area (Long)
Allows users to enter up to 131,072 characters on separate lines.
Text Area (Rich)
Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.
Text (Encrypted)
Allows users to enter any combination of letters and numbers and store them in encrypted form.
Time
Allows users to enter a local time. For example, "2:40 PM", "14:40", "14:40:00", and "14:40:50.608" are all valid times for this field.
URL
Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

11. Conclusion

The development and deployment of FitHub CRM represents a significant step forward in the digital transformation of fitness center management. By harnessing the power of the Salesforce platform — its custom object modeling, automation through Flows, role-based access control, and real-time analytics — the project demonstrates how a modern CRM system can fundamentally reshape the operational landscape of an industry that has traditionally relied on manual, fragmented processes.

The system successfully addresses every challenge identified in the problem statement. Member management is now centralized, accurate, and easily accessible to authorized staff. Trainer assignments and class scheduling are managed with conflict-detection mechanisms that prevent the double-booking errors that plagued the manual approach. Booking confirmations and payment reminders are dispatched automatically, reducing administrative workload and improving the member experience. The automated birthday communication flow adds a meaningful personalized touch that strengthens member loyalty without requiring any manual effort from staff.

The FitHub CRM system was completed across a structured six-sprint Agile development cycle, with each sprint delivering a tested and working increment of functionality. The final system passes all functional test cases — including record creation, validation rule enforcement, flow activation, email dispatch, and report generation — and demonstrates the performance characteristics required for production use.

From a learning perspective, this project provided extensive hands-on experience with the Salesforce development ecosystem, including custom object design, SOQL-driven formula field logic, declarative Flow Builder automation, and the Salesforce Lightning experience framework. These skills are directly applicable to a wide range of enterprise CRM roles, making FitHub CRM a project of significant professional value in addition to its academic merit.

12. Future Enhancements

While FitHub CRM in its current form delivers comprehensive fitness center management capabilities, several enhancements are planned for future development iterations. These enhancements would further increase the system's value to fitness businesses and extend its capabilities to address emerging operational needs.

12.1 Mobile CRM Integration

The current FitHub CRM implementation is primarily optimized for desktop browser usage. A future enhancement would involve building a dedicated Salesforce Mobile App configuration, allowing trainers to view their daily session schedules, mark attendance, and update member progress notes directly from a smartphone. Members could receive push notifications through the Salesforce Mobile Publisher app for upcoming class reminders and booking confirmations.

12.2 AI-Powered Member Retention Analytics

Salesforce's Einstein Analytics platform offers machine learning-powered predictive capabilities that could be integrated into FitHub CRM in a future release. An AI-driven churn prediction model could analyze member visit frequency, class booking patterns, payment history, and engagement metrics to identify members who are at risk of not renewing their membership. The system could then automatically trigger targeted retention campaigns — such as discount offers or complimentary personal training sessions — for at-risk members, enabling proactive intervention before the membership lapses.

12.3 WhatsApp and SMS Notification Integration

While the current system sends automated notifications via email, many members prefer to receive communications through messaging platforms such as WhatsApp or SMS. A future enhancement would integrate FitHub CRM with a third-party messaging API (such as Twilio or the WhatsApp Business API) using Salesforce's External Services framework. This would enable the system to send class reminders, booking confirmations, and birthday greetings directly to a member's mobile phone via their preferred messaging channel.

12.4 Inventory and Equipment Management

Fitness centers maintain a significant inventory of equipment — treadmills, dumbbells, resistance bands, yoga mats, and more — that requires regular maintenance and tracking. A future module could add an Equipment Inventory custom object to FitHub CRM, enabling staff to log equipment purchase dates, maintenance schedules, and service histories. Automated alerts could notify the facilities manager when equipment

maintenance is due or when inventory levels for consumables (such as cleaning supplies or paper towels) fall below a minimum threshold.

12.5 Multi-Branch Support

As a fitness chain expands to multiple locations, the CRM system must be capable of managing members, trainers, and classes across different branches while maintaining clear data segregation between locations. A future enhancement would introduce a Branch custom object linked to all other objects through a Lookup Relationship, and a role hierarchy and sharing rules structure that enforces branch-level data visibility — ensuring that a manager at Branch A can see only their branch's data while a corporate administrator retains cross-branch visibility.

12.6 Online Member Self-Service Portal

A Salesforce Experience Cloud (formerly Community Cloud) portal could be built to allow members to log in and manage their own accounts directly. Through this portal, members could view their booking history, schedule new class bookings, download payment receipts, update their contact information, and view their trainer's recommended workout plan. This self-service capability would significantly reduce the volume of routine administrative requests handled by front-desk staff, freeing them to focus on delivering a higher-quality in-person experience.

— End of Report —